

Automotive Skid Transfer Station

PLC software for an automotive skid transfer station connecting storage and post-storage conveyor lines.

Introduction

This project is inspired by a transfer station from an automotive production line where I currently work. It is not a copy of the original software. The entire implementation was developed from scratch for learning and professional development, using only the machine's mechanical operation and workflow as a reference.

In the real plant, this station is part of a much larger transport system controlled by a Siemens S7-400 PLC and several distributed I/O modules. The control architecture, software organization, and functional design differ from those used in this project.

To keep the project within a reasonable scope and complete it from start to finish, I decided to model only the transfer station and implement it on a Siemens S7-1200 CPU 1214C DC/DC/DC PLC as the main control platform. Since the number of input and output signals exceeds the CPU's built-in capacity, digital expansion modules were added to represent all the sensors and actuators of the station.

The application was developed entirely in TIA Portal V17 using a combination of Ladder (LAD) and Structured Control Language (SCL). Ladder was used to implement the sequential logic and actuator control, while SCL was mainly used for data copy operations, interface organization, and variable handling, where a textual programming language provided a clearer and more expressive solution.

The project also includes an HMI application developed for a Siemens TP700 Comfort panel, featuring automatic operation screens, manual and maintenance modes, process status visualization, and an alarm management system.

System Overview

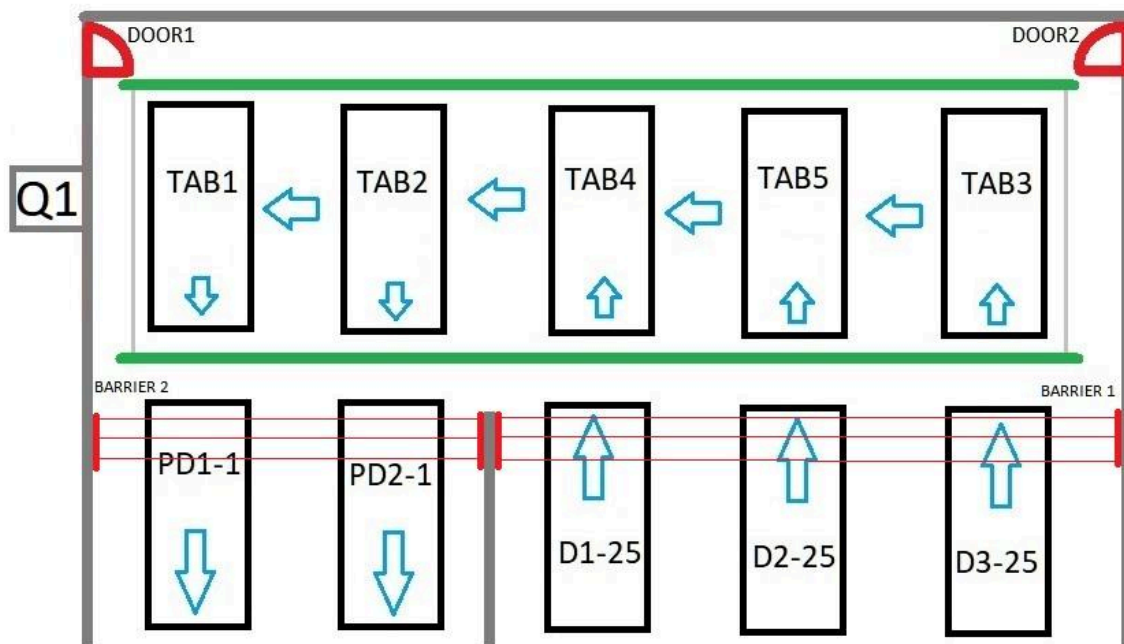


Figure 1: General layout of the transfer station.

The purpose of the developed station is to transfer car bodies mounted on skids between different storage areas using a chain conveyor system and lifting tables.

The system consists of a main chain conveyor and five lifting tables that transfer the skid between the different levels of the system. These tables operate together with three input storage areas (D1, D2, and D3) and two output storage areas (PD1 and PD2).

From a functional point of view, storage area D3 feeds only output storage area PD1, as both belong to the same production flow. In contrast, storage areas D1 and D2 both feed output storage area PD2. When both storage areas have a car body available, the system alternates between them to determine which one will perform the next transfer, ensuring a balanced production flow. If only one storage area contains a car body, the transfer is performed directly from that storage area.

The complete transfer sequence requires coordinating the vertical movement of the lifting tables with the horizontal movement of the chain conveyor, ensuring the correct position of the skid at all times and preventing interference between the different stages of the process.

Mechanical Layout

The following pages present the overall layout of the system and the simplified drawings used as a reference to understand the mechanical operation of the transfer station.

These diagrams show the arrangement of the storage areas, lifting tables, and chain conveyor, making it easier to understand the control sequences implemented later in the software.

The station includes five lifting tables, each with three lifting positions, and one chain conveyor on each side of the tables to move skids transversely between them.

It is important to understand the function of each table position. When a table is in the upper position, no transverse movement can take place because the skid is not in contact with the chain conveyor. This position is used for the *feeding* and *waiting* states.

The middle position is used only to receive a skid from an adjacent table. Each table is equipped with mechanical stoppers that prevent the skid from passing through while the table is in the middle position.

The lower position is used only to transfer a skid to an adjacent table.

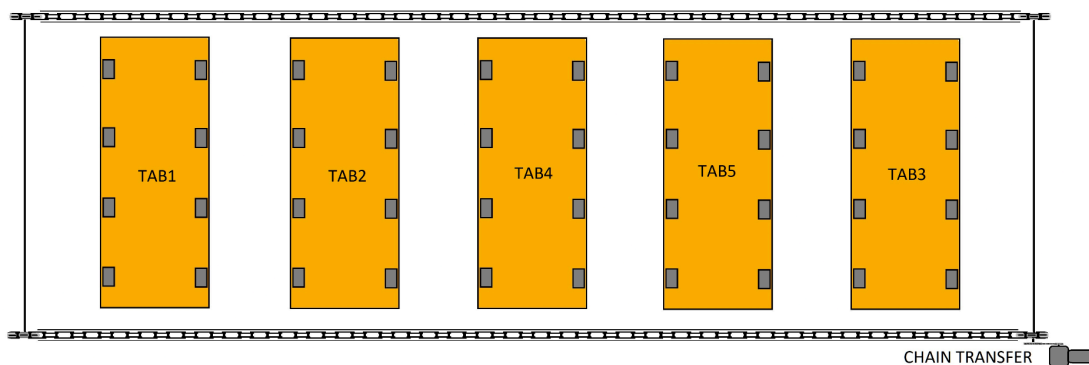


Figure 2: Mechanical Layout – Top View

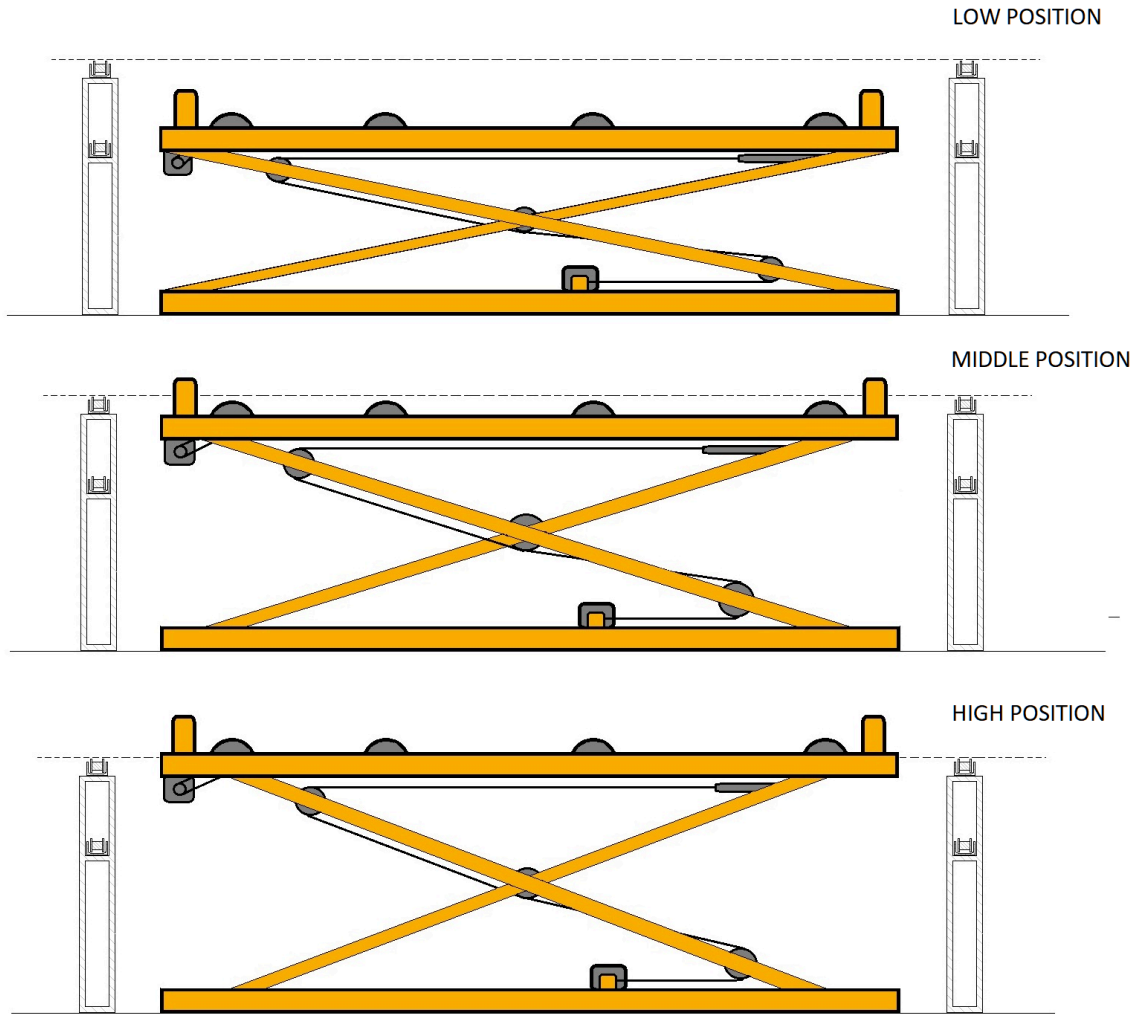


Figure 3: Cross-Sectional Side View of the Lifting Table in the Upper, Middle, and Lower Positions

Control Sequence

The control sequence was programmed so that whenever a position becomes available in “PostDeposit2”, the system requests a “feeding” transfer from either “Deposit1” or “Deposit2”, depending on the priority management. The same logic applies to “PostDeposit1”, which is supplied only by “Deposit3”.

During the “feeding” state, the corresponding lifting table is called and moved to its upper position. Once the skid reaches the table, the controller checks the status of the next table. If it is occupied, the sequence enters the “waiting” state. Otherwise, it switches to the “PassingNext” state.

During the “PassingNext” state, the current table moves down while the next table simultaneously moves to its middle position. At the same time, the chain conveyor motor is started to transfer the skid between the two tables.

After the skid reaches the next table, the controller evaluates the situation again and selects the next state: “waiting”, “PassingNext”, or “discharge”, depending on the skid position and the availability of the following station.

From a functional point of view, the state distribution is as follows:

- TAB5, TAB4, and TAB3 implement the “feeding”, “waiting”, and “PassingNext” states.
- TAB1 and TAB2 implement the “PassingNext”, “waiting”, and “discharge” states.

A photoelectric sensor installed on TAB2 detects whether a car body arriving from “Deposit3” must be routed to “PostDeposit1”. This is required because these car bodies have a longer rear section.

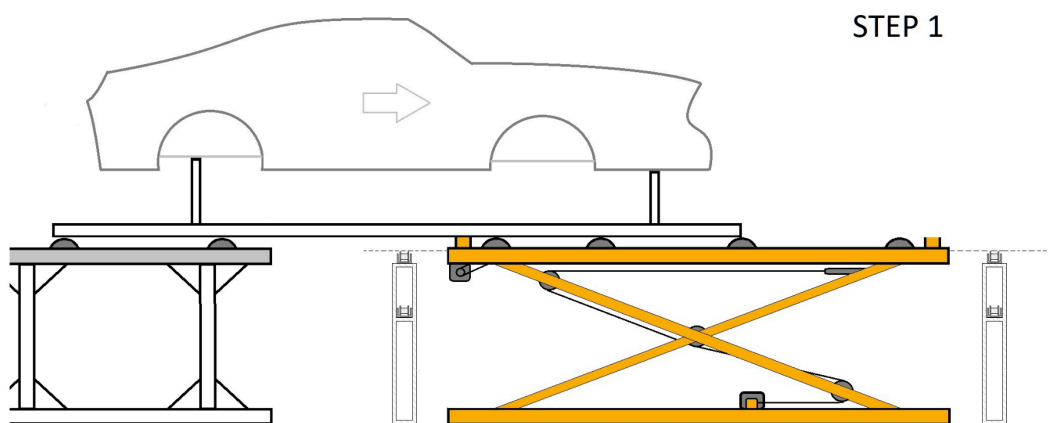


Figure 4: Feeding State with the Table in the Upper Position

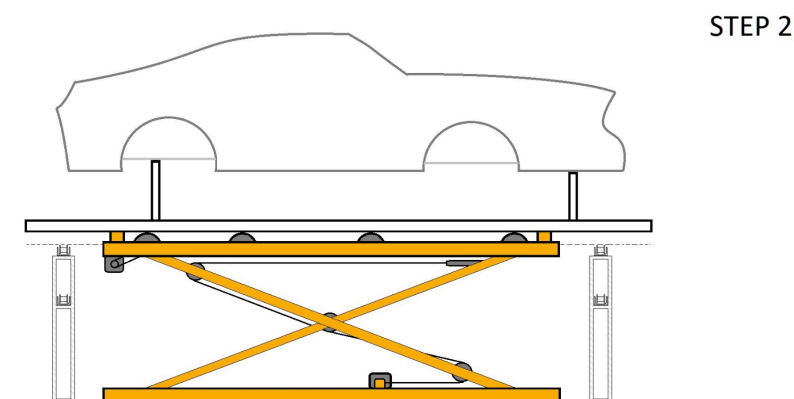


Figure 5: Skid Positioned in the “waiting” State

STEP 3

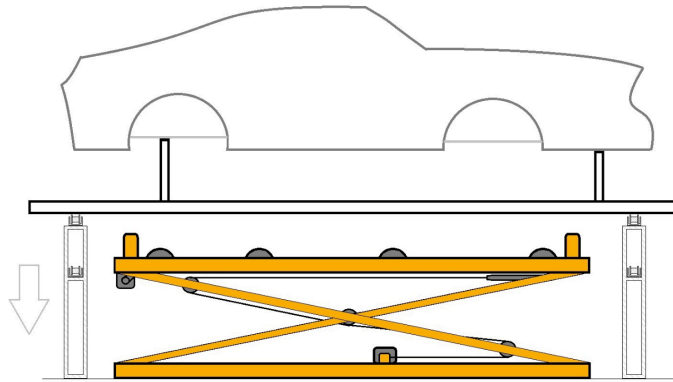


Figure 6: Start of the "PassingNext" State – Table Lowering

STEP 4

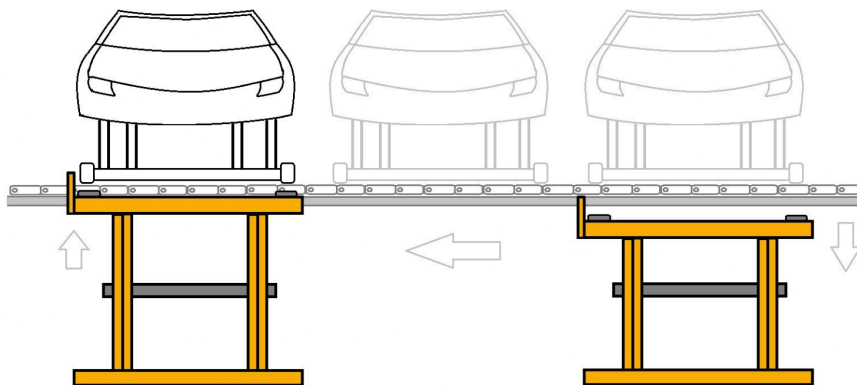


Figure 7: Skid Transfer on the Chain Conveyors to the Next Table in the Middle Position

STEP 5

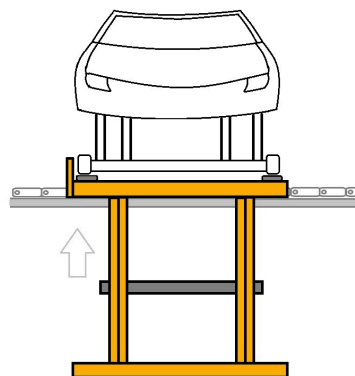
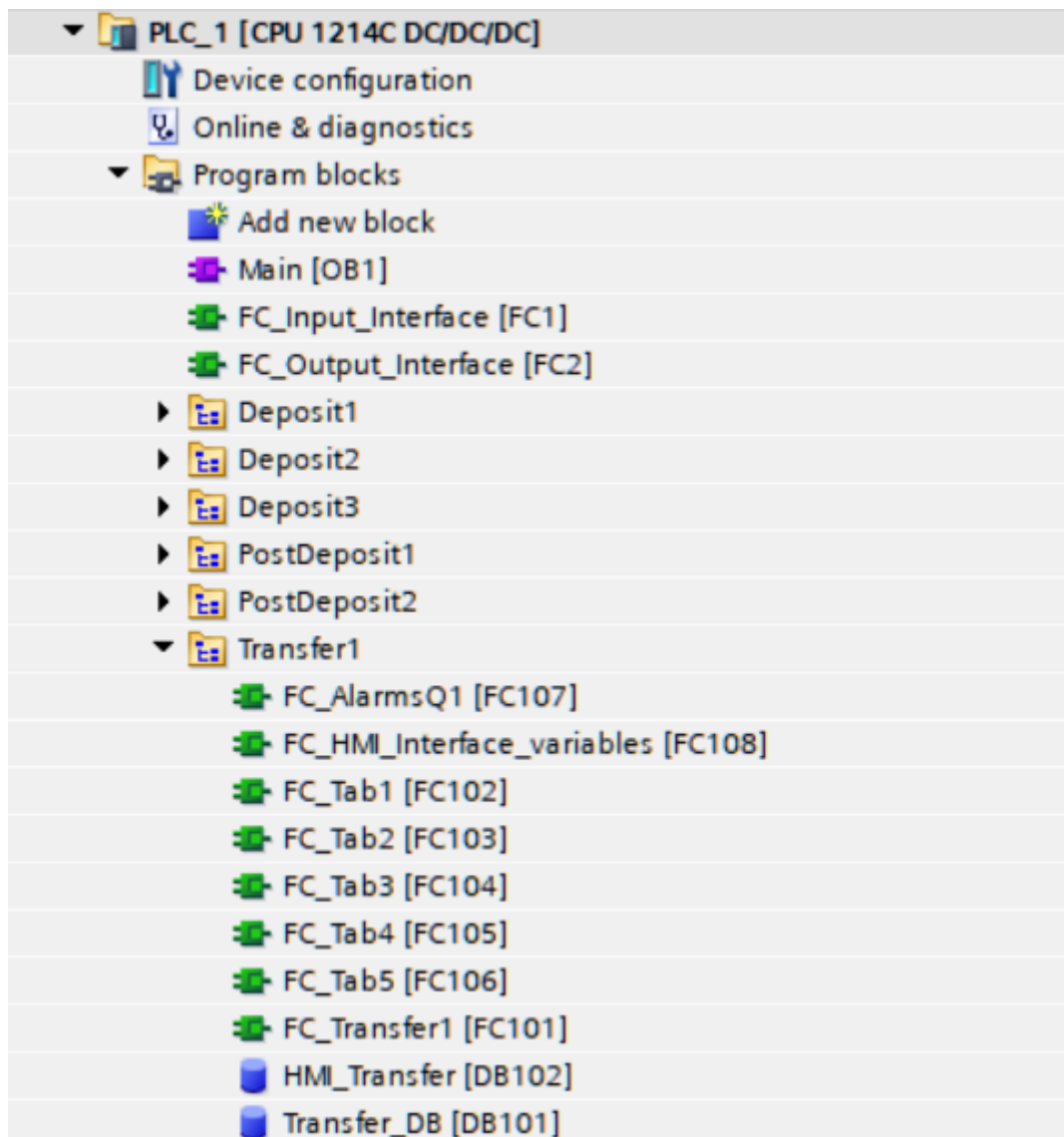


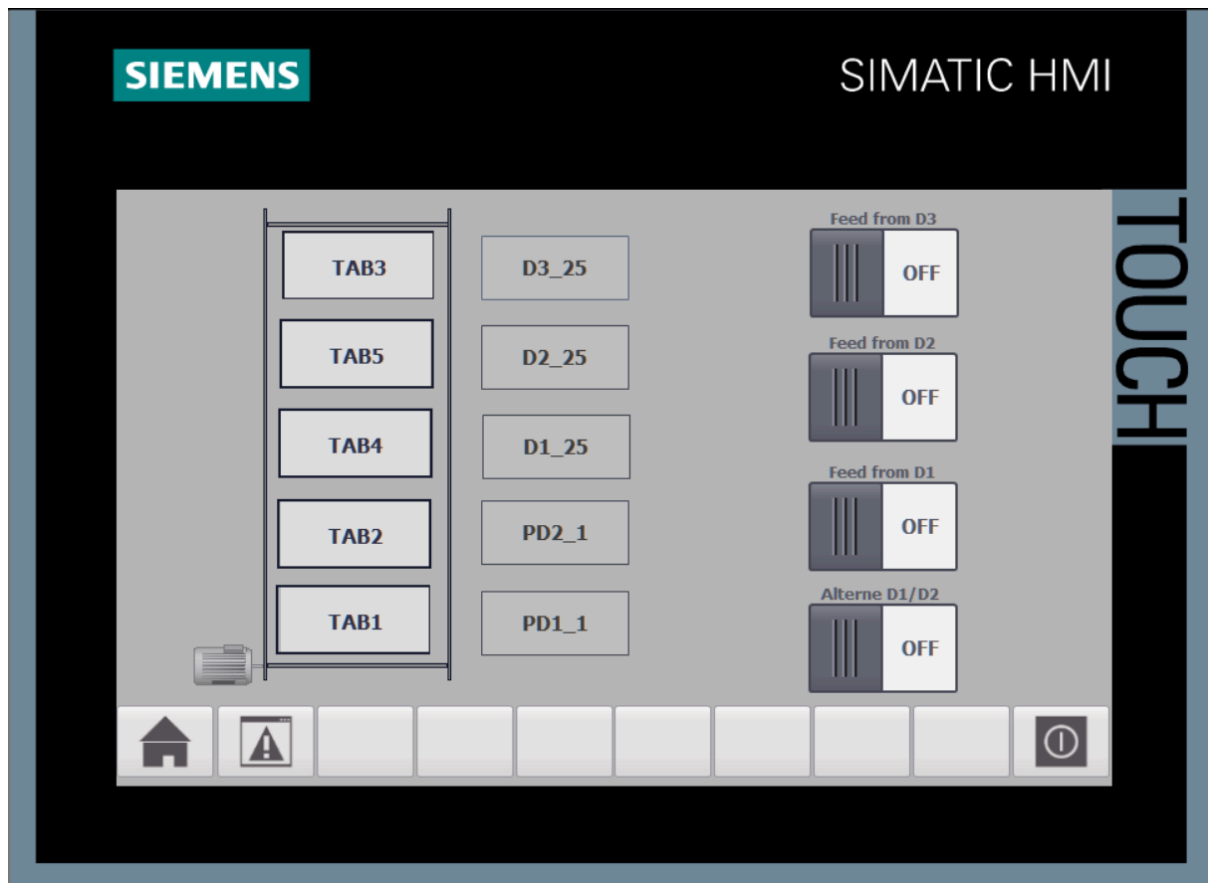
Figure 8: Skid Reached the Table and Raised to the Upper Position

Software Architecture



HMI

The project includes a minimalist HMI interface consisting of an overview screen displaying the system layout, an alarm management screen, and a dedicated screen for each lifting table. Each table screen provides the manual controls for the corresponding table, including the control of the chain conveyor motor.



Alarms Management

This project uses a "freeze actuators" philosophy whenever an error, alarm, or emergency condition is detected. In this situation, all actuators are frozen while the sequence states are preserved.

The control logic is designed to recover from the exact point where execution was interrupted, rather than resetting the sequence. This approach is necessary because the application contains multiple parallel states that depend on each other, making a complete reset neither practical nor desirable.

The safety functions are handled by a safety relay that monitors both safety doors and both safety light curtains. When all safety devices are in a safe state, the relay enables a safety output connected to a PLC input.

Manual Mode and Manual Maintenance Mode

The manual mode was designed to allow only movements that are valid for the current state of the system. For example, it is not possible to move the chain conveyors while a "feeding" state is active. This limitation prevents operations that could damage the machine or create unsafe situations, since manual mode may be used by any machine operator.

The maintenance manual mode works differently. In this mode, the state-based restrictions are bypassed, giving the operator full control over the actuators. For this reason, it is intended to be used only by qualified maintenance personnel who are familiar with the machine and its operation. Because all actuator movements are available, this mode requires special care during operation. Operation in this mode is limited only by the safety system. If the safety relay is activated, all actuator movement is stopped.

Both the manual mode and the maintenance manual mode are enabled by physical selector switches located on the electrical control panel. The maintenance manual mode selector is key-operated, and the key should remain under the control of maintenance personnel only.

Automotive Skid Transfer Station

PLC Software Design

Siemens S7-1200 TIA Portal V17

Author

A. Fernando Gola

2026